On Combiners for Verifiable Random Functions

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Cryptographic Hardness Assumptions

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Don’t bet on a single assumption.
Combine candidates based on different ones.

Research Question

Can we *combine* many verifiable random function (VRF) instances such that the resulting VRF is secure if *enough* candidates are secure?

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Can we *combine* verifiable random function (VRF) instances in a black-box fashion such that the resulting VRF is secure if *k*-out-of-*n* candidates are secure?

Verifiable Functions

Definition (Verifiable Function)

A *verifiable function* (VF) VF is a triple of the following efficient algorithms:

- $\mathsf{Gen}_{\mathsf{VF}}(1^\lambda) = (\mathsf{sk}, \mathsf{pk})$ generates a key-pair
- $\mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x) = (y, \pi)$ evaluates points to an image-proof pair
- $\mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk}, x, y, \pi) = b$ verifies evaluations

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A VF is a purely syntactic object, we define security properties on top of it.

Properties of Verifiable Functions

Correctness

For all $(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^{\lambda})$, $x \in \{0,1\}^{\lambda}$, and $(y, \pi) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x)$:

$$\mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk}, x, y, \pi) = 1.$$

Properties of Verifiable Functions

Unique Provability

For every public key pk and every point x, at most one image y is provable:

$$\mathrm{Vfy}_{\mathrm{VRF}}(\mathsf{pk}, x, y_0, \pi_0) = \mathrm{Vfy}_{\mathrm{VRF}}(\mathsf{pk}, x, y_1, \pi_1) = 1 \implies y_0 = y_1.$$

Properties of Verifiable Functions

(ε, t)-Pseudorandomness

For any adversary $\mathcal{A}$ running in time at most t:

$$\mathbf{Adv}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

Pseudorandom if $\varepsilon \in \mathsf{negl}(\lambda)$ for $t \in \mathsf{poly}(\lambda)$.

Pseudorandomness Experiment

Experiment $\mathbf{Exp}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A})$

1 : $b \leftarrow_{\$} \{0,1\}$
2 : $(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^{\lambda})$
3 : $S \leftarrow \varnothing$
4 : $(x^*, \sigma) \leftarrow_{\$} \mathcal{A}_1^{\mathcal{O}_{\mathsf{eval}}}(1^{\lambda}, \mathsf{pk})$
5 : $(y_0, \pi_0) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x^*)$
6 : $y_1 \leftarrow_{\$} \mathcal{Y}$
7 : $b' \leftarrow_{\$} \mathcal{A}_2(1^{\lambda}, x^*, \sigma, y_b)$
8 : **return** $b = b' \wedge x^* \notin S$

Oracle $\mathcal{O}_{\mathsf{eval}}(x)$

1 : $S \leftarrow S \cup \{x\}$
2 : $(y, \pi) \leftarrow_{\$} \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x)$
3 : **return** y

Properties of Verifiable Functions

(ε, t)-Weak Pseudorandomness

For any adversary $\mathcal{A}$ running in time at most t in the wprf game:

$$\mathbf{Adv}^{\mathrm{wprf}}_{\mathrm{VF}}(\lambda, \mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathrm{wprf}}_{\mathrm{VF}}(\lambda, \mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

Weakly pseudorandom if $\varepsilon \in \mathsf{negl}(\lambda)$ for $t \in \mathsf{poly}(\lambda)$.

Weak Pseudorandomness Experiment

Experiment $\mathbf{Exp}^{\mathsf{wprf}}_{\mathsf{VF}}(\lambda, \mathcal{A})$

1 : $b \leftarrow_{\$} \{0,1\}$
2 : $(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^{\lambda})$
3 : $S \leftarrow \varnothing$
4 : $\sigma \leftarrow_{\$} \mathcal{A}_1^{\mathcal{O}_{\mathsf{eval}}}(1^{\lambda}, \mathsf{pk})$
5 : $x^* \leftarrow_{\$} \{0,1\}^{\lambda}$
6 : $(y_0, \pi_0) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x^*)$
7 : $y_1 \leftarrow_{\$} \mathcal{Y}$
8 : $b' \leftarrow_{\$} \mathcal{A}_2(1^{\lambda}, x^*, \sigma, y_b)$
9 : **return** $b = b' \wedge x^* \notin S$

Oracle $\mathcal{O}_{\mathsf{eval}}(x)$

1 : $S \leftarrow S \cup \{x\}$
2 : $(y, \pi) \leftarrow_{\$} \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x)$
3 : **return** y

Properties of Verifiable Functions

(ε, t)-Unpredictability

For any adversary $\mathcal{A}$ running in time at most t in the up game:

$$\mathbf{Adv}^{\mathsf{up}}_{\mathsf{VF}}(\lambda, \mathcal{A}) := \Pr\left[\mathbf{Exp}^{\mathsf{up}}_{\mathsf{VF}}(\lambda, \mathcal{A}) = 1\right] \leq \varepsilon(\lambda)$$

Unpredictable if $\varepsilon \in \mathsf{negl}(\lambda)$ for $t \in \mathsf{poly}(\lambda)$.

Unpredictability Experiment

Experiment $\mathbf{Exp}^{\mathsf{up}}_{\mathsf{VF}}(\lambda, \mathcal{A})$

1 : $(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^{\lambda})$
2 : $S \leftarrow \varnothing$
3 : $(x^*, y^*) \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\mathsf{eval}}}(1^{\lambda}, \mathsf{pk})$
4 : $(y, \pi) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x^*)$
5 : **return** $y^* = y \wedge x^* \notin S$

Oracle $\mathcal{O}_{\mathsf{eval}}(x)$

1 : $S \leftarrow S \cup \{x\}$
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3 : **return** y

Verifiable Random Functions

Verifiable Random Function

A *verifiable random function* (VRF) is a VF that is:

- Correct
- Uniquely Provable
- Pseudorandom

Weak Verifiable Random Functions

Weak Verifiable Random Function

A *weak verifiable random function* (wVRF) is a VF that is:

- Correct
- Uniquely Provable
- Weakly Pseudorandom

Verifiable Unpredictable Functions

Verifiable Unpredictable Function

A *verifiable unpredictable function* (VUF) is a VF that is:

- Correct
- Uniquely Provable
- Unpredictable a

Why Should We Care About VRFs?

- Distributed Leader Election in PoS blockchains [KRDO17; DGKR18; BASV23]

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- Resettable Zero-Knowledge proofs [MR01]

Robust Black-Box Combiners

Definition (Robust $(n; k)$ Black-Box Combiner (Informal))

A robust $(n; k)$ black-box P-combiner Π_P is an efficient (PPT) oracle Turing machine such that:

- Given oracle access to n candidate schemes $\mathcal{O}_1, \ldots, \mathcal{O}_n$
- Implements primitive P
- Is secure if at least k candidates are secure

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- Implements primitive P
- Is secure if at least k candidates are secure **even against adversaries with a PSPACE oracle**

Example

Algorithm $\Pi_{\mathsf{PRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{PRF}}(k_1, k_2, x)$

1 : $y_1 \leftarrow \mathcal{O}_1.\mathsf{Eval}_{\mathsf{PRF}}(k_1, x)$
2 : $y_2 \leftarrow \mathcal{O}_2.\mathsf{Eval}_{\mathsf{PRF}}(k_2, x)$
3 : **return** $y_1 \oplus y_2$

Figure: Robust $(2; 1)$ Black-Box PRF-Combiner.

Robust Black-Box DAG Combiners

Definition (Robust $(n; k)$ Black-Box DAG Combiner, informal)

Algorithms of $\widetilde{\Pi}_{\mathsf{P}}$ are specified by:

- a DAG $\mathcal{G}$ with a source s and a sink t
- vertex labels $\ell : V(\mathcal{G}) \to [n] \times [|\mathsf{P}|]$
- a PPT machine f^v at each vertex v

Execution: traverse $\mathcal{G}$ in topological order. At vertex v, f^v computes a query for candidate $\ell(v)$.

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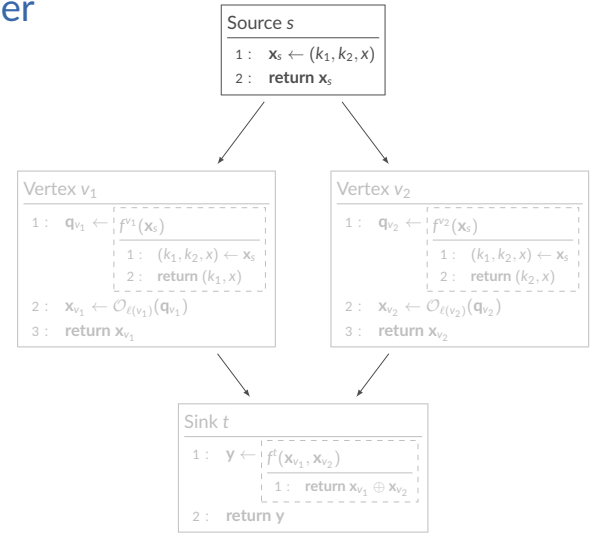
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Why DAGs? Decouples local computation from oracle queries.

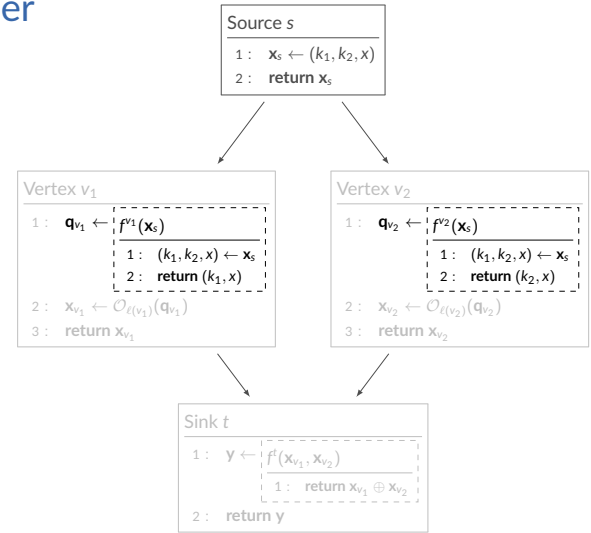
Example: DAG PRF-Combiner

- *Source s*: forwards (k_1, k_2, x).



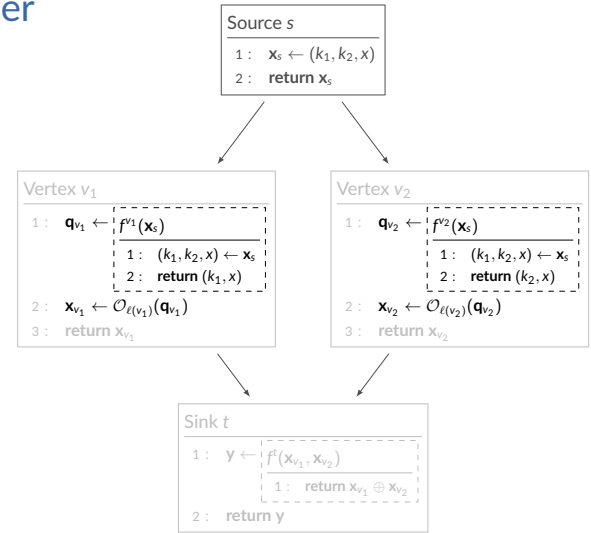
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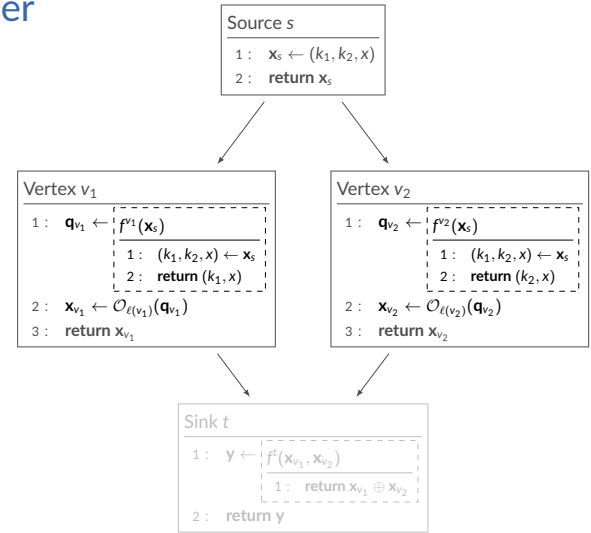
Example: DAG PRF-Combiner

- *Source s*: forwards (k_1, k_2, x).
- *Intermediate vertices*:
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  - query the candidate oracle $\mathcal{O}_{\ell(v)}$



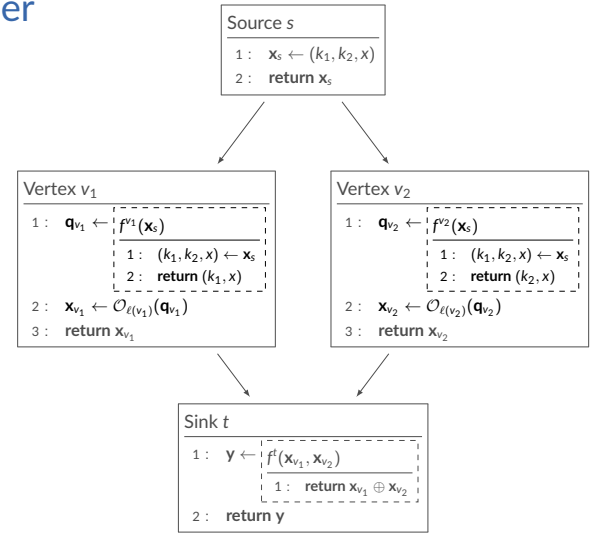
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- *Source s*: forwards (k_1, k_2, x).
- *Intermediate vertices*:
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- *Sink t*: computes the final output.



Generality of DAG Combiners



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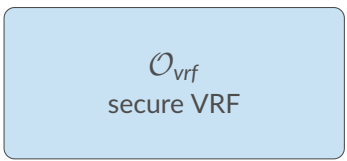
It suffices to (dis)prove the *DAG* version.

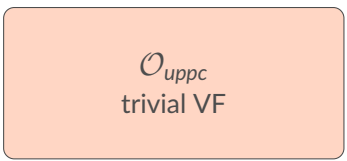
Main Theorem

There exists no robust $(2; 1)$ black-box VRF-combiner.

The Two Candidates

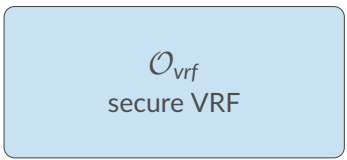
Suppose $\widetilde{\Pi}_{\mathsf{VRF}}$ is a robust $(2; 1)$ black-box DAG VRF-combiner.

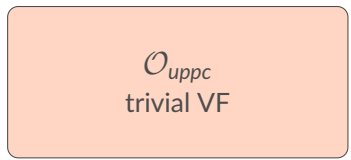




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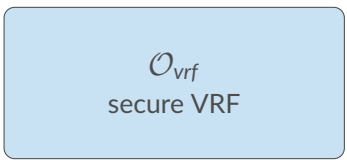


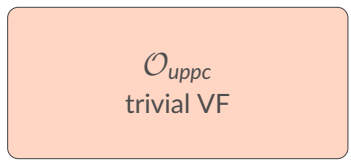


$\mathcal{O}_{uppc}$: every $\mathsf{Eval}(\mathsf{sk}, x)$ returns the same default y_{def}.

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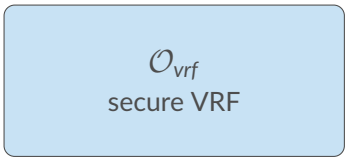


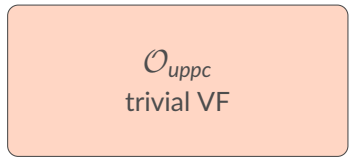
$\mathcal{O}_{uppc}$: every $\mathsf{Eval}(\mathsf{sk}, x)$ returns the same default y_{def}.

$\mathcal{O}_{vrf}$ secure $\Rightarrow$ $\widetilde{\Pi}_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{uppc}}$ is a secure VRF.

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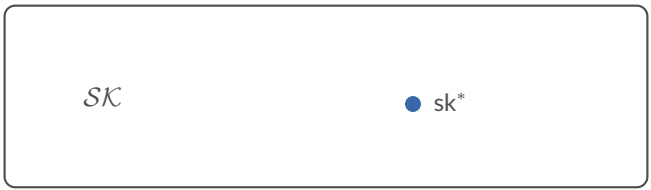
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Goal: adversary $\mathcal{A}^{\mathrm{PSPACE}}$ breaking its pseudorandomness.

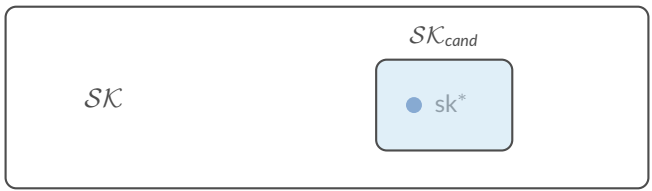
The Idea

The pseudorandomness experiment hides a secret key $\mathsf{sk}^* \in \mathcal{SK} = \{0,1\}^{s(\lambda)}$.



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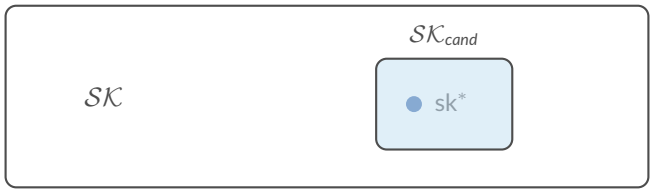
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Use the PSPACE oracle to *filter* $\mathcal{SK}$ down to a small $\mathcal{SK}_{cand}$ that still contains sk^*.

The Idea

The pseudorandomness experiment hides a secret key $\mathsf{sk}^* \in \mathcal{SK} = \{0,1\}^{s(\lambda)}$.



Use the PSPACE oracle to *filter* $\mathcal{SK}$ down to a small $\mathcal{SK}_{cand}$ that still contains sk^*.
If $|\mathcal{SK}_{cand}| \in \mathsf{poly}$, sampling uniformly gives $\mathsf{sk}_{cand} = \mathsf{sk}^*$ with non-negligible probability.

$\mathsf{EvalSim}_{\mathsf{VRF}}$ as a Black Box

Assume an algorithm $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x, y, \pi, \tau) \in \{\top, \bot\}$ with the following properties:

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- returns $\top$ on the experiment's secret key sk^*

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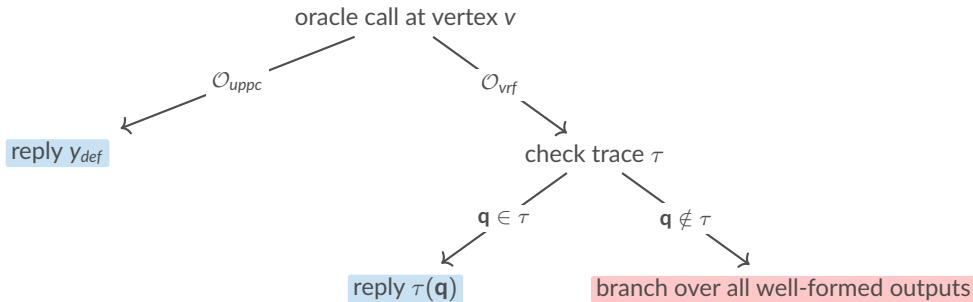
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Constructing $\mathsf{EvalSim}_{\mathsf{VRF}}$ *and proving these properties hold is the technical core of the thesis.*

Constructing $\mathsf{EvalSim}_{\mathsf{VRF}}$

Traverse the DAG in topological order. At each oracle-call vertex:



Return $\top$ iff *every* branch produces the honest image y.

$\mathsf{EvalSim}_{\mathsf{VRF}}$ Revisited

Properties of $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x, y, \pi, \tau)$:

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3. Compute the verification traces $\tau_i \leftarrow \mathsf{trace}\big(\widetilde{\Pi}_{\mathsf{VRF}}^{\mathcal{O}_{vrf}, \mathcal{O}_{uppc}}.\mathsf{Vfy}(\mathsf{pk}^*, x_i, y_i, \pi_i)\big)$.

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4. Use PSPACE to sample $\mathsf{sk}_{cand} \leftarrow_\$ \mathcal{SK}_{cand}$, where

$$\mathcal{SK}_{cand} := \big\{\mathsf{sk} \in \mathcal{SK} : \mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x_i, y_i, \pi_i, \tau_i) = \top \ \forall i\big\}.$$

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1. Sample $x_1, \ldots, x_{s+\lambda} \leftarrow_\$ \{0,1\}^\lambda$ (recall $\mathcal{SK} = \{0,1\}^{s(\lambda)}$).
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3. Compute the verification traces $\tau_i \leftarrow \mathsf{trace}\big(\widetilde{\Pi}_{\mathsf{VRF}}^{\mathcal{O}_{vrf}, \mathcal{O}_{uppc}}.\mathsf{Vfy}(\mathsf{pk}^*, x_i, y_i, \pi_i)\big)$.
4. Use PSPACE to sample $\mathsf{sk}_{cand} \leftarrow_\$ \mathcal{SK}_{cand}$, where
$$\mathcal{SK}_{cand} := \big\{\mathsf{sk} \in \mathcal{SK} : \mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x_i, y_i, \pi_i, \tau_i) = \top \ \forall i\big\}.$$
5. Sample challenge $x^* \leftarrow_\$ \{0,1\}^\lambda$ and obtain y^*.

The Adversary

1. Sample $x_1, \ldots, x_{s+\lambda} \leftarrow_\$ \{0,1\}^\lambda$ (recall $\mathcal{SK} = \{0,1\}^{s(\lambda)}$).
2. Query the eval oracle: $(y_i, \pi_i) \leftarrow \mathcal{O}_{\mathsf{eval}}(x_i)$.
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5. Sample challenge $x^* \leftarrow_\$ \{0,1\}^\lambda$ and obtain y^*.
6. Check $\widetilde{\Pi}_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{uppc}}.\mathsf{Eval}(\mathsf{sk}_{cand}, x^*) \stackrel{?}{=} (y^*, \pi)$.

Bounding the Advantage

$$\mathbf{Adv}^{\mathsf{prf}}_{\widetilde{\Pi}^{\mathcal{O}_{vrf},\mathcal{O}_{uppc}}_{\mathsf{VRF}}}(\lambda,\mathcal{A}) = 2 \cdot \left|\Pr[\mathcal{A}\text{ wins}] - \tfrac{1}{2}\right|$$

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$$\mathbf{Adv}^{\mathsf{prf}}_{\widetilde{\Pi}^{\mathcal{O}_{vrf},\mathcal{O}_{uppc}}_{\mathsf{VRF}}}(\lambda,\mathcal{A}) \notin \mathsf{negl}(\lambda) \;\Rightarrow\; \text{contradicts pseudorandomness.} \quad \square$$

Results

Theorem

No robust $(2; 1)$ black-box VRF-combiner.

Corollary

No robust $(n; \lfloor n/2 \rfloor)$ black-box VRF-combiner.

Conjecture

No robust $(n; n - 1)$ black-box VRF-combiner.

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No robust $(2; 1)$ black-box VRF/VUF/wVRF-combiner.

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Contraposition $\Rightarrow$ no $(n; \lfloor n/2 \rfloor)$ black-box VRF-combiner.

Beyond Impossibility

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a simple robust $(n; 1)$ black-box VRF-combiner exists.

Robust $(n; 1)$ Black-Box UP-VRF-Combiner

$\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x)$

1 : $(\mathsf{sk}_1, \mathsf{sk}_2) \leftarrow \mathsf{sk}$
2 : $(y_1, \pi_1) \leftarrow \mathcal{O}_1.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_1, x)$
3 : $(y_2, \pi_2) \leftarrow \mathcal{O}_2.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_2, x)$
4 : $y \leftarrow y_1 \oplus y_2$
5 : $\pi \leftarrow (y_1, \pi_1, y_2, \pi_2)$
6 : **return** (y, π)

$\Pi_{\mathsf{VRF}}^{\mathcal{O}_1,\mathcal{O}_2}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi)$

1 : $(\mathsf{pk}_1, \mathsf{pk}_2) \leftarrow \mathsf{pk}$
2 : $(y_1, \pi_1, y_2, \pi_2) \leftarrow \pi$
3 : $b_1 \leftarrow \mathcal{O}_1.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}_1, x, y_1, \pi_1)$
4 : $b_2 \leftarrow \mathcal{O}_2.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}_2, x, y_2, \pi_2)$
5 : **return** $b_1 \wedge b_2 \wedge (y = y_1 \oplus y_2)$

Robust $(n; 1)$ Black-Box UP-VRF-Combiner

$\Pi^{\mathcal{O}_1,\mathcal{O}_2}_{\mathsf{VRF}}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x)$

1 : $(\mathsf{sk}_1, \mathsf{sk}_2) \leftarrow \mathsf{sk}$
2 : $(y_1, \pi_1) \leftarrow \mathcal{O}_1.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_1, x)$
3 : $(y_2, \pi_2) \leftarrow \mathcal{O}_2.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_2, x)$
4 : $y \leftarrow y_1 \oplus y_2$
5 : $\pi \leftarrow (y_1, \pi_1, y_2, \pi_2)$
6 : **return** (y, π)

$\Pi^{\mathcal{O}_1,\mathcal{O}_2}_{\mathsf{VRF}}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi)$

1 : $(\mathsf{pk}_1, \mathsf{pk}_2) \leftarrow \mathsf{pk}$
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4 : $b_2 \leftarrow \mathcal{O}_2.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}_2, x, y_2, \pi_2)$
5 : **return** $b_1 \wedge b_2 \wedge (y = y_1 \oplus y_2)$

Iterate pairwise to obtain a robust $(n; 1)$ combiner.

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Summary

- Introduced Robust $(n; k)$ Black-Box DAG Combiner for more principled analysis
- Established impossibility for robust $(n; \lfloor n/2 \rfloor)$ black-box VRF-combiner
- Conjectured impossibility for robust $(n; n-1)$ black-box VRF-combiner
- Constructed robust $(n; 1)$ black-box VRF-combiner given u.p. and p.c. candidates

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